2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
−1	1	−2	4	−1	2
0	0	−1	1	−2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2 = 10$	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

$\bar{x}=1$ $\bar{y}=2$

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

b.

$$b_2 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} = \frac{8}{10} = 0.8$$ when x increase 1 unit, y increase 0.8 unit

$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8\cdot 1 = 1.2$ when $x=0$, $y=1.2$

c

$\sum_{i=1}^{n} x_i^2 = 9+4+1+1+0 = 15$ $\sum_{i=1}^{n} x_i y_i = 12+4+3-1 = 18$

$\sum x_i^2 - N\bar{x}^2 = 15 - 5\cdot 1 = 10 = \sum(x_i-\bar{x})^2$

$\sum x_i y_i - N\bar{x}\bar{y} = 18 - 5\cdot 1\cdot 2 = 8 = \sum(x_i-\bar{x})(y_i-\bar{y})$

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

$\hat{y} = 1.2 + 0.8x$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	−0.8	0.64	−1.6
1	3	2	1	1	1
−1	1	0.4	0.6	0.36	−0.6
0	0	1.2	−1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i\hat{e}_i = 0$

$\sum(y_i-\bar{y})^2 = 4+0+1+1+4 = 10$

$\sum(x_i-\bar{x})^2 = 10$

$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

$$s_y^2 = \frac{\sum(y_i-\bar{y})^2}{N-1} = \frac{10}{4} = 2.5$$

$$s_x^2 = \frac{\sum(x_i-\bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

$CV_x = 100\left(\frac{s_x}{\bar{x}}\right) = 100\left(\frac{1.5811}{1}\right) \doteq 158.11\%$

$s_x = s_y = \sqrt{2.5} \doteq 1.5811$

median$(x) = 1$

$$s_{xy} = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{N-1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5}\sqrt{2.5}} = 0.8$$

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

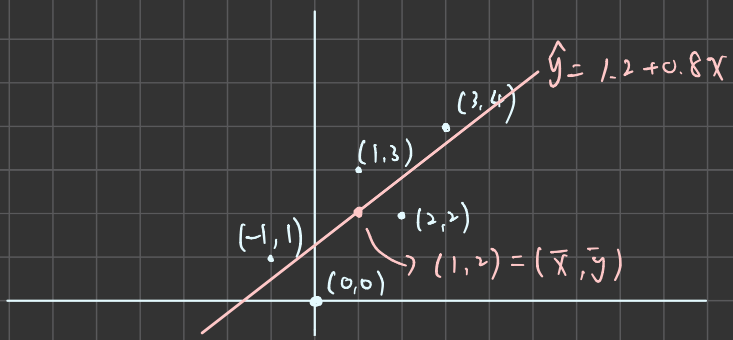
h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i / N$.

i. Compute $\hat{\sigma}^2$.

j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$.

e.

f.



g. $b_1 + b_2\bar{x} = 1.2 + 0.8(1) = 2$

h. $\sum \hat{y}_i = 10 \quad \bar{\hat{y}} = \frac{10}{5} = 2 = \bar{y}$

i. $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

j. $\widehat{Var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum(x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{0.12} \doteq 0.3464$

hw01

2026-03-05

＃2-22

```
star5_small <- read.csv("/Users/chenjunqi/Downloads/star5_small.csv")
data_ab <- subset(star5_small, aide == 0)
```

#a

```
#綜合分數 (TOTALSCORE) 對 小班虛擬變數 (SMALL)
mod_a <- lm(totalscore ~ small, data = data_ab)
summary(mod_a)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = data_ab)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -189.44  -50.44   -9.44   43.06  324.38
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.675 249.396   <2e-16 ***
## small         12.175      5.369   2.268   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.59 on 773 degrees of freedom
## Multiple R-squared:  0.006608,   Adjusted R-squared:  0.005323
## F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

totalscore=916.412+12.175small 當學生在常規班級時，學生的平均綜合性向測驗分數為 916.442 分。 被分配到小班的學生，其平均綜合分數會比常規班的學生高出 12.175 分。

#b

```
mod_b_math <- lm(mathscore ~ small, data = data_ab)
summary(mod_b_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = data_ab)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -144.777  -35.028   -5.028   29.223  142.223
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.449 197.545   <2e-16 ***
## small          5.251      3.578   1.467    0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.71 on 773 degrees of freedom
## Multiple R-squared:  0.002778,   Adjusted R-squared:  0.001488
## F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

mathscore=483.777+5.251small 當學生在常規班級時，平均的數學測驗分數為 483.777 分。 被分配到小班的學生，其平均數學分數比常規班的學生高出 5.251 分。

```
# 篩選資料：排除小班 (SMALL == 1)，剩下的就是常規班(0)與助理常規班(1)
data_cd <- subset(star5_small, small == 0)
```

#c迴歸模型：綜合分數 (TOTALSCORE) 對 助理虛擬變數 (AIDE)

```
mod_c <- lm(totalscore ~ aide, data = data_cd)
summary(mod_c)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = data_cd)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.748  -50.442   -5.442   40.558  312.558
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.559 257.529   <2e-16 ***
## aide           4.306      4.994   0.862    0.389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.23 on 835 degrees of freedom
## Multiple R-squared:  0.0008898,  Adjusted R-squared:  -0.0003068
## F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

當學生在沒有配置助理的常規班級，學生的平均綜合性向測驗分數為 916.442 分。 班上多了一位全職教師助理，學生的平均綜合分數會比一般常規班高出 4.306 分。 p-value > 0.05 不顯著，在常規班級中增加一名全職教師助理，對學生的綜合學習效果沒有顯著的幫助。

#d迴歸模型：閱讀 (READSCORE) 與 數學 (MATHSCORE) 對 助理虛擬變數 (AIDE)

```
mod_d_read <- lm(readscore ~ aide, data = data_cd)
summary(mod_d_read)
```

```
##
## Call:
## lm(formula = readscore ~ aide, data = data_cd)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -74.665 -21.536  -2.536  15.464 194.335
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.460 296.341   <2e-16 ***
## aide           2.871      2.049   1.401    0.161
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 29.64 on 835 degrees of freedom
## Multiple R-squared:  0.002347,   Adjusted R-squared:  0.001152
## F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

```
mod_d_math <- lm(mathscore ~ aide, data = data_cd)
summary(mod_d_math)
```

```
##
## Call:
## lm(formula = mathscore ~ aide, data = data_cd)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -146.212  -31.212   -5.777   29.223  142.223
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.355 205.464   <2e-16 ***
## aide           1.435      3.304   0.434    0.664
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.79 on 835 degrees of freedom
## Multiple R-squared:  0.0002258,  Adjusted R-squared:  -0.0009715
## F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

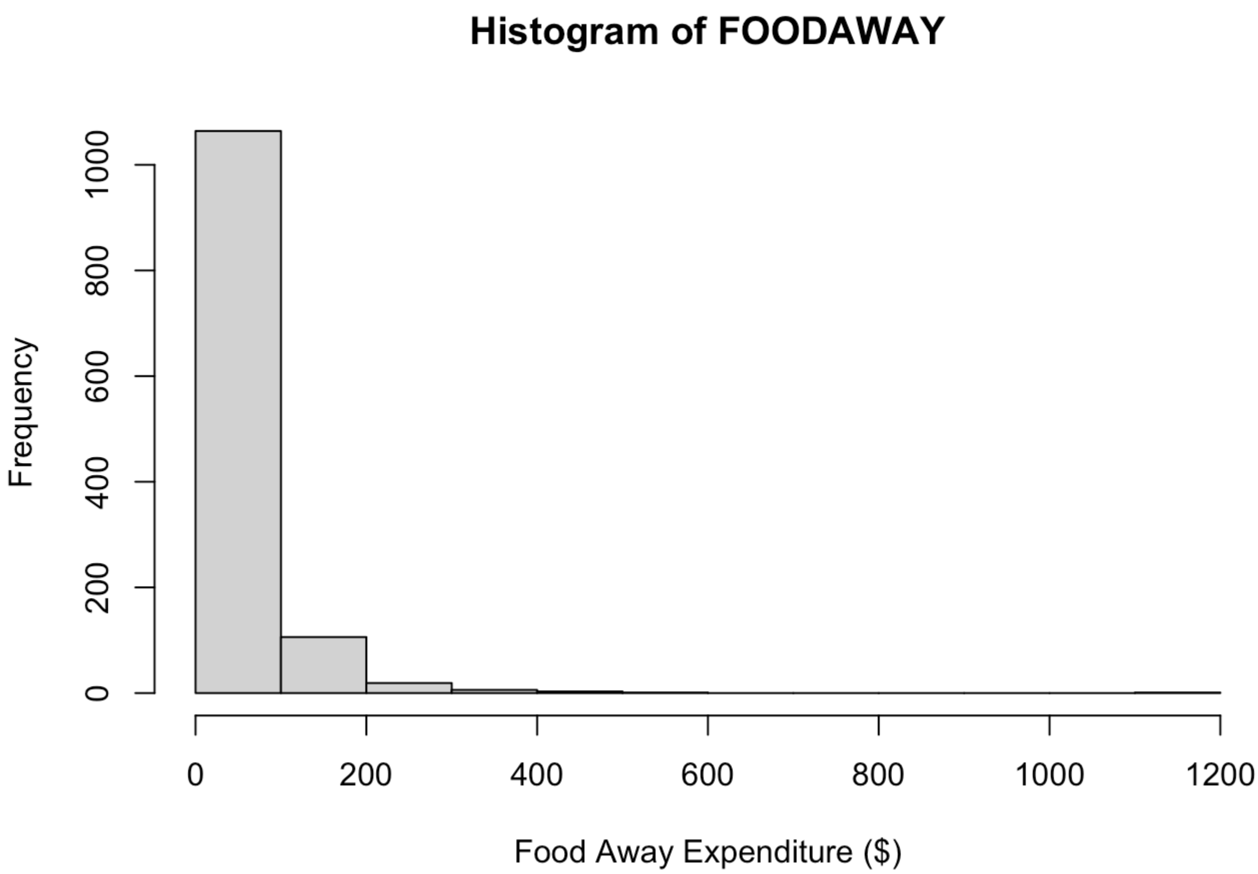
兩者皆不顯著，增加助理對提升閱讀和數學成績沒有顯著幫助。

2-25

```
cex5_small <- read.csv("/Users/chenjunqi/Downloads/poe5csv/cex5_small.csv")
```

#a

```r
# 畫出直方圖
hist(cex5_small$foodaway, main="Histogram of FOODAWAY", xlab="Food Away Expenditure ($)")
```




```r
# 查看基礎敘述統計 (這會顯示 Mean, Median, 1st Qu.(25%), 3rd Qu.(75%))
summary(cex5_small$foodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   12.04   32.55   49.27   67.50 1179.00
```

Mean：49.27 Median：32.55 25th percentile：12.04 75th percentile：67.50

#b

```r
# 有進階學位 (Advanced degree)
adv_data <- subset(cex5_small, advanced == 1)
mean(adv_data$foodaway)
```

```
## [1] 73.15494
```

```r
median(adv_data$foodaway)
```

```
## [1] 48.15
```

```
# 有大學學位 (College degree，且排除有進階學位的)
col_data <- subset(cex5_small, college == 1 & advanced == 0)
mean(col_data$foodaway)
```

```
## [1] 48.59718
```

```
median(col_data$foodaway)
```

```
## [1] 36.11
```

```
# 兩者皆無 (No advanced or college degree)
none_data <- subset(cex5_small, college == 0 & advanced == 0)
mean(none_data$foodaway)
```

```
## [1] 39.01017
```

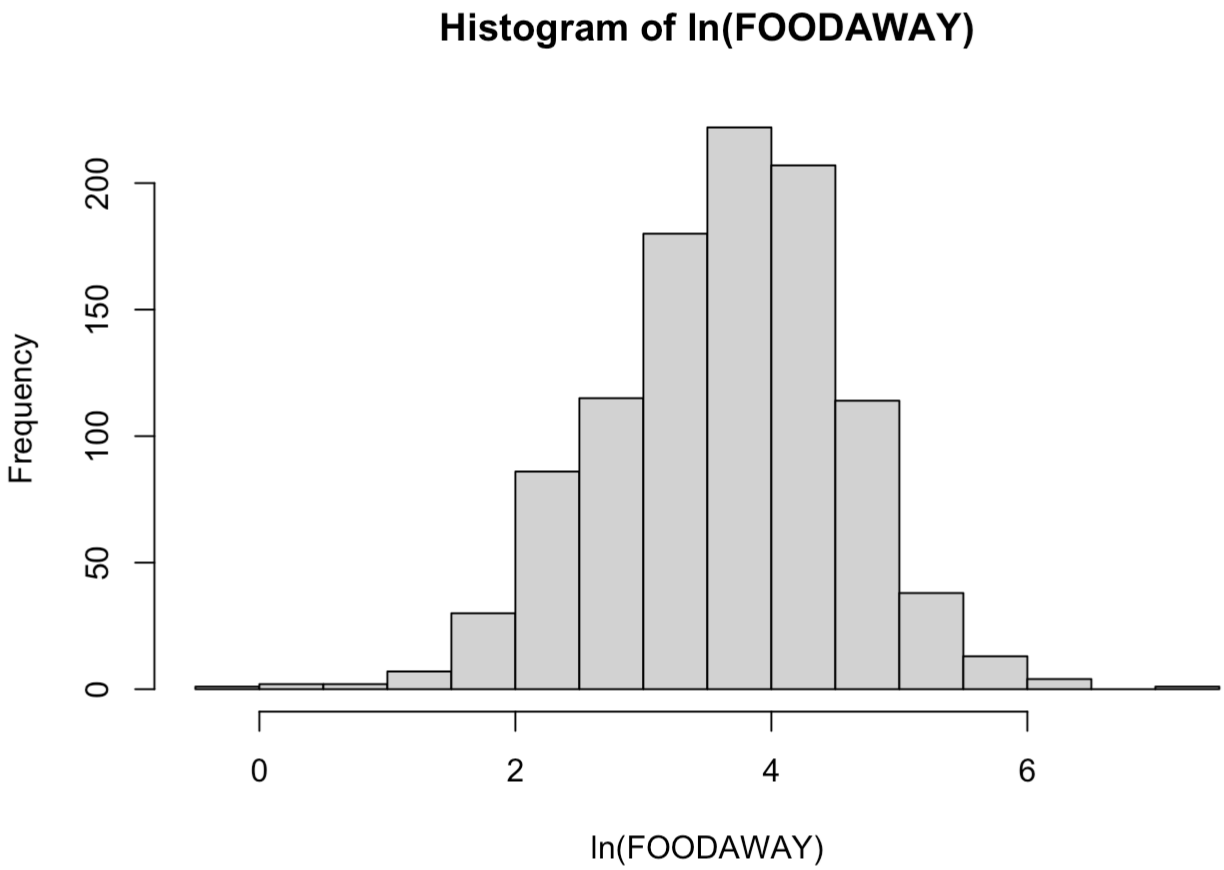
```
median(none_data$foodaway)
```

```
## [1] 26.02
```

#c

```
# 取自然對數
cex_clean <- subset(cex5_small, foodaway > 0)
```

```
# 畫直方圖與看敘述統計
hist(log(cex_clean$foodaway), main="Histogram of ln(FOODAWAY)", xlab="ln(FOODAWAY)")
```




```
summary(log(cex_clean$foodaway))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
## -0.3011  3.0759  3.6865  3.6508  4.2797  7.0724
```

因為有些家庭在外用餐的支出為0。ln(0)會趨近於負無限大。因此當對這個變數取對數時，那些支出為 0 的觀察值會被系統視為無效值而自動剔除，導致總觀察值數量減少。

#d

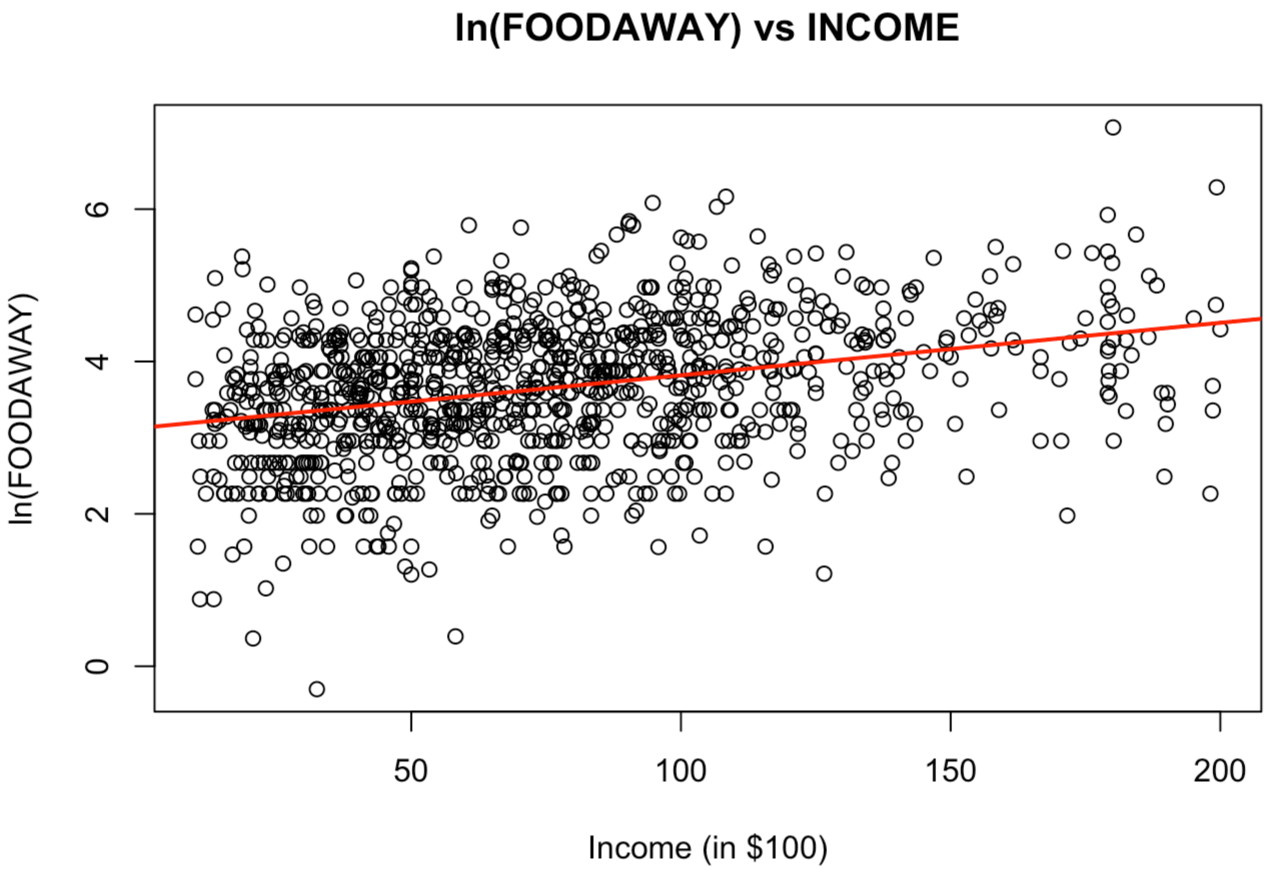
```
# 使用剛剛過濾掉 0 的乾淨資料來跑迴歸
mod_d <- lm(log(foodaway) ~ income, data = cex_clean)
summary(mod_d)
```

```
##
## Call:
## lm(formula = log(foodaway) ~ income, data = cex_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.6547 -0.5777  0.0530  0.5937  2.7000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
## income      0.0069017  0.0006546   10.54   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8761 on 1020 degrees of freedom
## Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
## F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

當家庭月收入增加1個單位（100 美元）時，在外用餐支出FOODAWAY預期將會改變大約 $(\beta_2 \times 100)\%$。

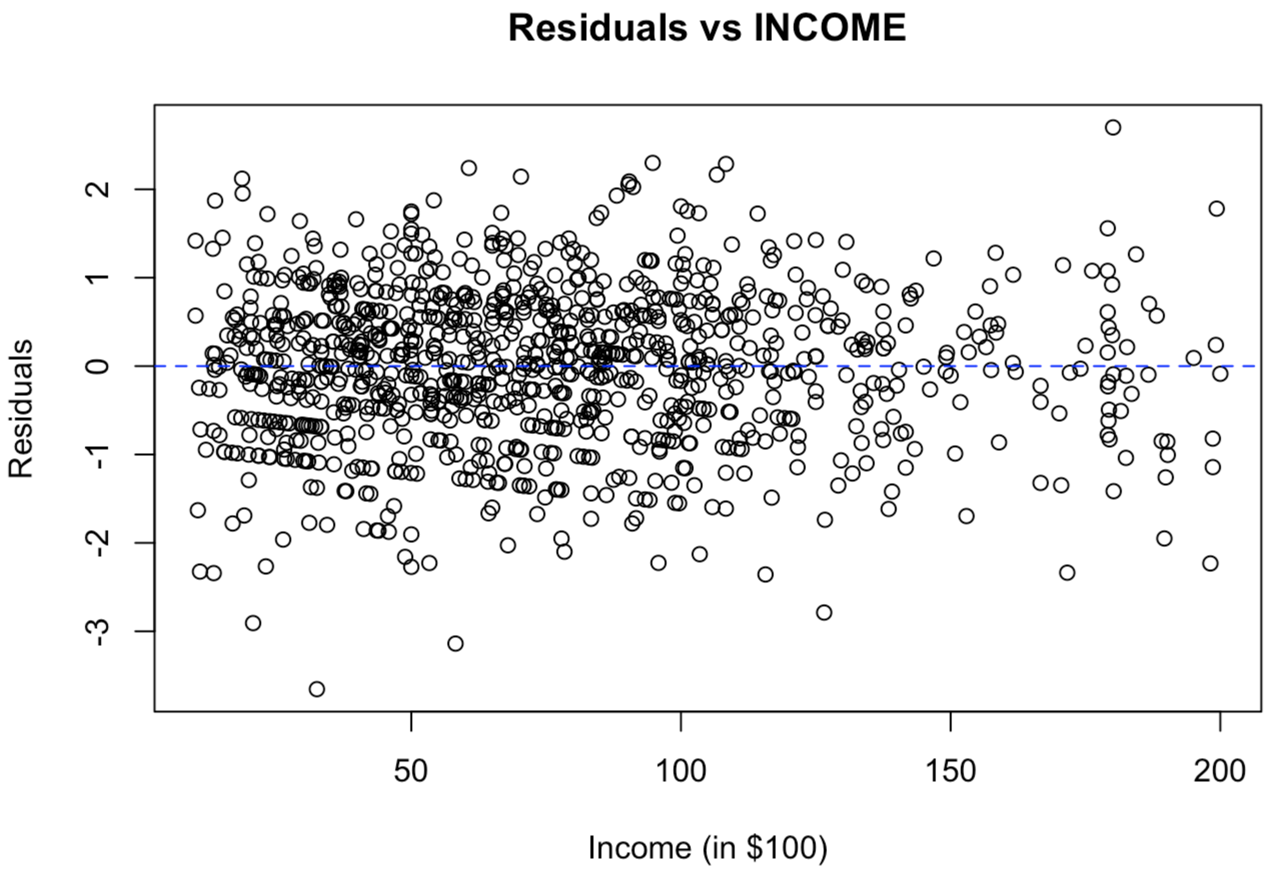
#e

```
# 畫出散佈圖
plot(cex_clean$income, log(cex_clean$foodaway),
     main="ln(FOODAWAY) vs INCOME",
     xlab="Income (in $100)",
     ylab="ln(FOODAWAY)")

# 加上迴歸線
abline(mod_d, col="red", lwd=2)
```



#f

```
# 取得殘差
res <- residuals(mod_d)

# 將殘差對 INCOME 作圖
plot(cex_clean$income, res,
     main="Residuals vs INCOME",
     xlab="Income (in $100)",
     ylab="Residuals")
abline(h=0, col="blue", lty=2) # 畫一條 y=0 的水平參考線
```



殘差平均分布在藍色虛線上下方，並無隨著x大小改變